

Multiple Choice (10 marks)

1. Calculate the length of a half-wave dipole antenna so as to optimally radiate a signal of 300 MHz when the velocity factor for the antenna element is (A) 0.70 (B) 1.0
 - (a) 0.7 meters, 1 meter
 - (b) 0.6 meters, 0.9 meters
 - (c) 0.5 meters, 1 meter
 - (d) 0.55 meters, 0.85 meters
 - (e) 0.25 meters, 0.75 meters
2. The frequency range of a commercially broadcast FM signal is 88 to 108 MHz, with carrier swing of 125 kHz. Find the percentage modulation of the signal.
 - (a) 110%
 - (b) 125%
 - (c) 50%
 - (d) 62.5%
 - (e) 75%
3. Find curl at the point (1, -2, 1) for $\mathbf{a} = x^2y^2\mathbf{i} + 2xyz\mathbf{j} + z^2\mathbf{k}$
 - (a) (4, 4, 0)
 - (b) (-2, 0, 0)
 - (c) (0, 0, 4)
 - (d) (4, 0, 0)
 - (e) (2, 0, 0)
4. Which of the following windings are necessary in case of all dc machines?
 - (a) star winding
 - (b) delta winding
 - (c) closed winding
 - (d) parallel winding
 - (e) zigzag winding
5. The current through a resistor of 2Ω is given by $i(t) = \cos^2 t$. Find the energy dissipated in the resistor from $t = 0$ to $t = 5$ sec.
 - (a) 5.75 J

- (b) 7.50 J
- (c) 4.75 J
- (d) 1.25 J
- (e) 3.75 J

True / False (10 marks)

Answer True or False

6. True or False: The lowest natural frequency for an automobile engine spring made of spring steel is 10 Hz.
7. True or False: The equivalent resistance of five resistors in parallel with values of 10, 15, 20, 25, and 30 Ω is 5 Ω .
8. True or False: The information in the message about the dice sum of seven is exactly 1 bit, indicating the probability of the sum being seven is 1/6.
9. True or False: A box that describes the effect of inputs on a control subsystem is known as a process block.
10. True or False: The coefficient of compressibility at constant temperature for an ideal gas is given by $\beta = 1/P$.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. In the combustion of propane with 125% theoretical air, calculate the dew point and mole fractions of the products at 1 atm pressure. Discuss the implications of these values in combustion efficiency.
12. Derive the homogeneous solution for the voltage $v_C(t)$ across a $(1/2)$ F capacitor in a series RLC circuit with $R = 4$, $L = 4$ H, $i_L(0) = (1/4)$ A, and $v_C(0) = 0$. Discuss your findings.

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